

数学作业纸

科目_____

班级:_____

姓名:_____

编号:_____

第____页

7.3 b 1. $R_A = \frac{39L}{8}$ $R_B = \frac{9L}{8}$

$$M(x) = \frac{39L}{8}x - \frac{1}{2}x^2 + \frac{1}{2}\left(x - \frac{L}{2}\right)^2$$

$$EI w'' = \frac{39L}{8}x - \frac{1}{2}x^2 + \frac{1}{2}\left(x - \frac{L}{2}\right)^2$$

$$EI \theta = EI w' = \frac{39L}{16}x^2 - \frac{1}{6}x^3 + \frac{1}{6}\left(x - \frac{L}{2}\right)^3 + C$$

$$EI w = \frac{39L}{48}x^3 - \frac{1}{24}x^4 + \frac{1}{24}\left(x - \frac{L}{2}\right)^4 + C(x + D)$$

$x=0$ 时, $w=0 \Rightarrow D=0$

$x=L$ 时, $w=0 \Rightarrow \frac{39L^4}{48} - \frac{1}{24}L^4 + \frac{1}{24}\left(\frac{L}{2}\right)^4 + CL = 0$

$$\frac{39L^4}{48} - \frac{1}{24}L^4 + \frac{1}{384}L^4 + CL = 0 \Rightarrow \frac{99L^4}{384} + CL = 0 \Rightarrow C = -\frac{39L^3}{128}$$

$$\theta_A = \frac{C}{EI} = -\frac{39L^3}{128EI} \quad (v)$$

$$\theta_B = \frac{1}{EI} \left(\frac{39L^3}{16} - \frac{1}{6}L^3 + \frac{1}{48}L^3 - \frac{39L^3}{128} \right) = \frac{79L^3}{384} \quad (v)$$

$$w_C = \frac{1}{EI} \left[\frac{39L}{16} \left(\frac{L}{2}\right)^3 - \frac{1}{24} \left(\frac{L}{2}\right)^4 - \frac{39L^3}{128} \left(\frac{L}{2}\right) \right] = -\frac{59L^4}{768} \quad (d)$$

$\frac{1}{2}L \leq x \leq \frac{3}{2}L$ 时 $\theta(x) = 0$

$$\frac{39L}{16}x^2 - \frac{1}{6}x^3 - \frac{39L^3}{128} = 0 \quad x \approx 0, 4598L$$

$$w_{max} = -\frac{5.049L^4}{768EI} \quad (d)$$

7.4 b.

$$w_{A1} = -\frac{F\left(\frac{1}{2}\right)^3}{3EI} = -\frac{FL^3}{24EI} \quad \theta_{A1} = -\frac{F\left(\frac{1}{2}\right)^2}{2EI} = -\frac{FL^2}{8EI}$$

$$\theta_{B1} = \theta_{A1} = -\frac{FL^2}{8EI} \quad w_{B1} = w_{A1} + \theta_{A1} \cdot \frac{1}{2} = -\frac{FL^3}{24EI} - \frac{FL^2}{8EI} \cdot \frac{1}{2} = -\frac{5FL^3}{48EI}$$

$$w_{A2} = \frac{(-F)L\left(\frac{1}{2}\right)^2}{2EI} = -\frac{FL^3}{8EI} \quad w_{B2} = \frac{(-F)L^2}{2EI} = -\frac{FL^3}{2EI} = -\frac{24FL^3}{48EI}$$

$$\theta_{B2} = \frac{(-F)L}{EI} = -\frac{FL^2}{EI} = -\frac{8FL^2}{8EI}$$

$$w_A = w_{A1} + w_{A2} = -\frac{FL^3}{24EI} - \frac{FL^3}{8EI} = -\frac{FL^3}{6EI} \quad (d)$$

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第____页

$$w_B = w_{B1} + w_{B2} = -\frac{FL^3}{48EI} - \frac{2FL^3}{48EI} = -\frac{3FL^3}{48EI} \quad (1)$$

$$\theta_B = \theta_{B1} + \theta_{B2} = -\frac{FL^2}{8EI} - \frac{8FL^2}{8EI} = -\frac{9FL^2}{8EI} \quad (2)$$